

UGTS LIBRARY EDITION

VOLUME 3

IDENTITY, RULES, MEMORY AND REPLAY

How a world knows what changed and why

Stable IDs, definitions, proposals, evidence, checkpoints and order

IDENTITY

Position, asset, node, route
and lineage are different.

DEFINITION

Rules have IDs, types,
dependencies and hashes.

PROPOSAL

A suggestion cannot mutate
authority.

COMMIT

Verified changes are ordered
and atomic.

REPLAY

Checkpoints and events
rebuild or expose divergence.

EVIDENCE

Observation, uncertainty and
permission remain explicit.

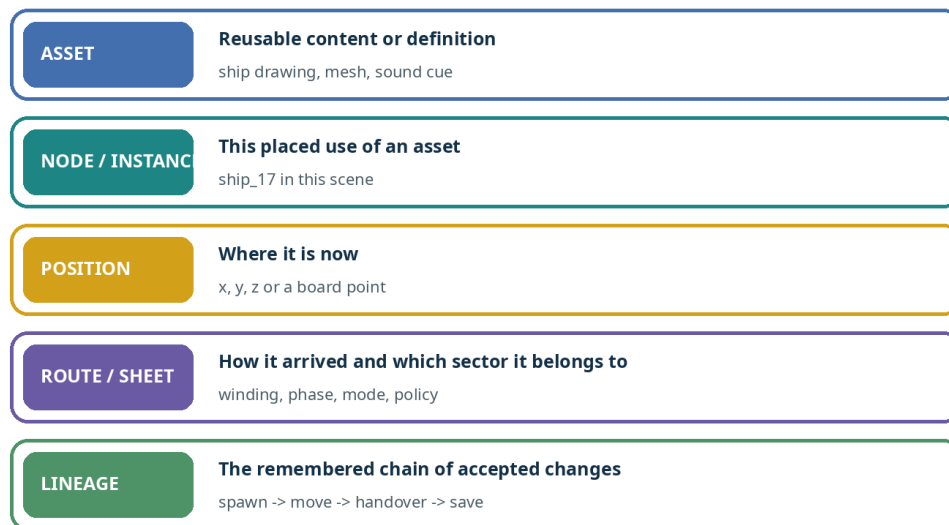
SAME PLACE DOES NOT MEAN SAME THING

Identity needs more than a coordinate

Two red books can sit on one shelf. Two game entities can overlap. A route can cross itself. A map observation can revisit the same doorway. The coordinate helps locate them, but stable identity, type, route and history say which thing or state sector is meant.

ONE THING CAN HAVE MANY DIFFERENT KINDS OF IDENTITY

Keeping these layers separate prevents “same picture” or “same position” from becoming a false identity claim.



A content hash can detect changed content. It does not replace the full record, the route history or legal identity.

CHILD SUMMARY

Where it is, what reusable thing it uses, which placed object it is, how it arrived and what happened to it are different questions.

STORY: THE TWO RED BOOKS

A library example of stable identity

Book A and Book B have the same red cover and are returned to the same shelf position. Book A has a repaired page and a waiting reservation. Book B is a different copy with another barcode and no reservation.

ASSET	Both may use the same cover design and edition definition.
NODE	Each physical copy has its own stable library item ID.
POSITION	Both may occupy the same shelf slot at different times, or even touch one another.
POLICY	The reservation applies to Book A, not every red book at that coordinate.
CHANGE	Returning, repairing or borrowing one copy changes that item record.
LINEAGE	The event trail preserves which copy moved, when, why and under which rule.

FORMAL LESSON

asset identity != node identity; node coordinate != lineage identity; content hash != legal or personal identity

ASSET, NODE, OBJECT, POSITION, ROUTE, LINEAGE

A builder's identity dictionary

LAYER	CONTRACT
definition / asset	Reusable content or rule family. Multiple instances may reference it.
node / instance	Stable placed identity in a scene, map, game or ledger.
object / entity	Runtime state bundle that may own components and behavior.
position	Current coordinate or cell; never complete identity by itself.
route / sheet / mode	State sector and compatibility context that can distinguish co-located states.
lineage	Accepted transition history, ancestry, handover, winding and external novelty.
hash	Compact integrity or divergence check over a canonical record; not the full semantic or legal identity.

DESIGN RULE

Store event-critical state compactly, but leave large meshes, source definitions, material graphs and editor metadata in referenced cold storage. Compact does not mean incomplete without a reconstruction contract.

Source note: Sources: KC Two Hands 3.0 state/identity split and asset/node contract; UGTS-KC 2.0 route identity; SCLP finite-key boundaries.

THE RULES MOVE INSIDE THE WORLD

UGTS-KC 3.6 makes definitions first-class substrate data

CHILD ROUTE

A rule card is no longer only a sentence outside the system. It receives a stable name, a type, inputs, outputs, dependencies, an order phase, a source note and a content check.

FIELD	MEANING
id	Stable definition name.
kind	What family of rule or object this is.
domain / codomain	What may go in and what kind of result may come out.
parameters	Declared literal settings.
dependencies	Other definitions required first.
phase	When it is evaluated.
invariants	What must remain true.
provenance	Source-derived, engineering-derived, corrected, bounded or other declared class.
content hash	Canonical change-detection value over the record.

REFERENTIAL CLOSURE

Unknown references, duplicate IDs, bad hashes and undeclared dependency cycles are rejected before execution.

DEPENDENCIES AND ORDER

A rule may need another rule first

Suppose “open the gate” depends on a gate definition, a key definition, a match operator and a time guard. The dependency graph tells us what must be resolved and constructed before the event can be checked.

DEPENDENCY ORDER Definitions form a directed graph. A deterministic topological order resolves prerequisites.	COMPOSITION ORDER H2 after H1 is not generally the same as H1 after H2. Transform order is stored literally.
PHASE ORDER Pure evaluation, certification, support, compatibility, guard, proposal and commit are different phases.	REPLAY ORDER Accepted events must be applied in their recorded sequence unless a versioned merge law says otherwise.
CYCLE POLICY A dependency cycle is rejected unless a fixed-point or recursion profile explicitly states initialization, convergence, depth and work budgets.	

Source note: Sources: UGTS-KC 3.6 dependency DAG and ten-phase schedule; UGTS-KC 4.2 distinguishes dependency, phase, composition, reduction, parallel, event and replay order.

WHAT A CONTENT HASH CAN AND CANNOT SAY

Useful integrity evidence with a narrow meaning

CAN HELP WITH

Detect changed canonical bytes, support cache keys, compare checkpoints, verify a definition record or find replay divergence.

CANNOT PROVE

Mathematical truth, human identity, ownership, authorization, truthful place/time, uncompromised hardware, safety or legal chain of custody.

Three rules

1. Hash the declared canonical record and version the canonicalization rules.
2. Keep the full state, definition or event record that gives the digest meaning.
3. Use independent provenance, authorization, signatures or measurements when the application requires them.

CHILD ANALOGY

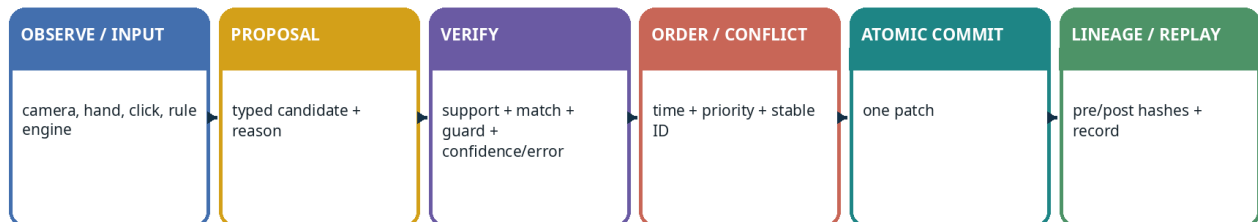
A tamper seal can show that a box changed. It cannot tell who owns the box or whether the contents are true.

PROPOSAL, VERIFICATION AND ATOMIC COMMIT

Parallel detection; one authoritative boundary

SUGGEST FIRST. CHECK. THEN CHANGE ONCE.

Sensors, hands, games and solvers may propose. Only the ordered verifier commits authority.



Projection, rendering or a model score stays downstream. It cannot silently become a committed fact.

WHY PROPOSALS MATTER

A camera, GPU, hand tracker, game system, search algorithm or human input may be useful and still be uncertain or conflicting. Proposal-only design keeps detection separate from authoritative mutation.

VERIFIED CONDITION

in support AND compatible AND accepted guard status AND confidence $\geq$ floor AND numeric error $\leq$ event margin AND all application policy gates pass

CONFLICTS AND SIMULTANEOUS EVENTS

A deterministic system must say what happens when patches compete

CASE	EXPLICIT POLICY
different fields	Both patches may be accepted if their declared effects are compatible.
same field, same value	A versioned commutative/idempotent merge may coalesce them.
same field, different values	Use priority and stable tie order, or emit a conflict event.
overlapping time intervals	Use certified event-time intervals and declare whether they are orderable.
coincident constraint	Enter a maintained constraint mode instead of emitting infinite zero-crossing events.
too dense / chatter	Use hysteresis, hold time, coalescing and a stop budget.

NEVER HIDE THE LOSERS

Rejected proposals keep reason codes such as `outside_support`, `incompatible`, `bad_guard`, `low_confidence`, `excessive_error` or `patch_conflict`. Diagnostics are part of explainability.

Source note: Sources: Two Hands 3.0 deterministic ordering/conflicts; KC Atlas spatial proposals; UGTS-KC 4.2 event and replay order.

CHECKPOINTS, HASHES AND REPLAY

Rebuild the story or stop at the first mismatch

REPLAY IS A CHECKABLE STORY OF ACCEPTED CHANGES

A checkpoint stores a trusted state. Later events rebuild the same ending or reveal the first mismatch.



If a pre-hash or post-hash differs, stop at the first divergence. A digest is a check, not a substitute for full state.

CHECKPOINT

A canonical snapshot at a known sequence. It shortens replay and provides a comparison anchor.

EVENT LOG

Ordered accepted patches with pre/post hashes, reason, route/lineage and external novelty where needed.

RECONSTRUCTIBILITY RULE

Closed deterministic behavior may be rebuilt from seed, definitions and events. External inputs, measurement evidence, topology edits, solver choices and calibration changes must be stored or checkpointed when they cannot be regenerated.

OBSERVATION IS NOT AUTHORITY

The central lesson of KC Atlas and the spatial branch

A camera, model or human note may observe a doorway, obstacle or crack. The observation carries support, source, calibration, uncertainty and confidence. It may propose a map patch, but it cannot directly rewrite the authoritative ledger.

CAPTURE	Record when, where, how and with which calibration profile the evidence was acquired.
OBSERVE	Create a typed observation with bounded uncertainty and provenance.
SUPPORT	Check whether it applies to the relevant map region, object and time.
VERIFY	Check compatibility, guard/error interval, stable identity and policy.
COMMIT	Apply an accepted node, edge or measurement patch with reason and hashes.
REPLAY	Rebuild the ledger, route query and change report from the accepted event sequence.
SAFETY LIMIT A synthetic route demo, image-only classification or reference ledger is not a real-world evacuation, structural, medical or legal certification.	

Source note: Sources: KC Atlas 3.9.3 and UGTS-KC 4.0 Spatial Evidence Ledger.

STABLE IDS AND REPEAT-SCAN CHANGE

Compare the same modeled thing, not merely nearby pixels

CHILD ROUTE

To say “this doorway changed,” the system must first keep a stable doorway identity. Otherwise it may confuse movement, a new camera angle, a different object or failed re-identification with a real change.

CONDITION	TRUSTWORTHY RESULT
same stable ID + non-overlapping measurement intervals	candidate real change
same stable ID + overlapping intervals	within tolerance / not confirmed
different or uncertain identity	re-identification problem, not a certified change
missing calibration or scale	measurement-critical claim deferred
conflicting sources	merge policy or unresolved evidence event required

ROUTE CONSEQUENCE

After a verified edge-state change, constrained reachability may change for one policy mode but remain available for another. The query result is only as trustworthy as the observations, identity and route policy.

SEED PLUS MEASURED EVIDENCE

KSEED stores a recipe and what the recipe cannot recreate

THE SEED CAN RECREATE

Deterministic schedules, stable synthetic IDs, sampling choices, fixtures and procedural display decisions.

THE FILE MUST STILL STORE

Measured luma/IMU summaries, feature and map evidence, accepted events, uncertainty, calibration, checksums and lineage.

KSEED 4.1 integrity layers

- Versioned header with profile commitments and CRC32.
- Typed chunk headers with sequence, lengths, record count, schema ID and CRCs.
- Conditional zlib only when it actually makes the stored chunk smaller.
- SHA-256 chain over ordered chunk headers and stored payloads.
- Replay stops on framing, sequence, CRC, decompression, chain or summary failure.

NO SEED-ONLY MAGIC

A seed does not reconstruct unstored photons. A nominal raw-input/evidence ratio is not equal-information image compression.

Source note: Sources: UGTS-KC 4.1 Spatial Seed Native and 4.1.1 Atlas Spatial Seed SLAM Fusion.

UGTS-KC 4.2: ORDER BECOMES DATA

The latest general substrate release found in the library audit

UGTS-KC 4.2 MAKES MANY KINDS OF ORDER EXPLICIT

The child six-question path remains inside a longer fifteen-phase technical schedule.



Order is data. An operator cannot silently choose its own precedence, commute an effect, change a reduction tree or turn a projection into truth.

STRONGEST UPGRADE

Ambiguity becomes representable and rejectable. A symbol cannot silently choose its domain; a transform cannot silently commute; a reduction cannot silently change trees; a score cannot silently become state; a projection cannot silently become truth.

FALLBACK POLICY

When a mature domain implementation is simpler, faster, safer or more accurate at equal error, use it. The general layer can remain a schema, oracle and event-handoff contract rather than replacing the domain runtime.

PROJECTION IS NOT TRUTH

Pictures, exports and actuators are optional downstream endpoints

OUTPUT	BOUNDARY
SVG / Canvas / Bayer pixels	Presentation or regression artifact; replaceable without changing authority.
glTF / USDA / mesh	Derived geometry or interchange; may carry independent approximation budgets.
camera preview / overlay	Operator aid; raw or processed image remains observation evidence, not map authority.
solver hint / model score	Proposal or diagnostic unless an application guard promotes a verified result.
actuator command	Downstream effect requiring device feedback, safety and calibration contracts.
content hash	Integrity/checkpoint signal; not proof of truth or ownership.

DELETE TEST

A presentation layer should be replaceable or removable without changing authoritative substrate records. When deleting a picture changes the world state, the architecture has likely mixed projection with authority.

PUBLIC INFORMATION IS NOT PERMISSION

An older-learner safety lesson from SARA

CORE LESSON

Seeing a public identifier, address, document or checksum does not grant permission to recover, use, alter or transfer its secret or protected counterpart.

ALLOWED EDUCATIONAL IDEAS

Public format validation, public test vectors, watch-only consistency, self-owned known-input reconstruction, test networks and non-enumerating search-space measurement.

REJECTED SCOPE

Third-party secret search, mnemonic/passphrase enumeration, private-key scans, signing, broadcast, balance targeting or funds transfer.

CERTIFICATE BOUNDARY

A checksum can detect some transcription errors. It does not authenticate a person, prove ownership or prove that an account has funds.

Source note: Source: UGTS-KC 3.6.3 SARA. This page teaches permission and secret-egress boundaries; it is not a children's wallet-building mechanic.

FOUR ACTIVITIES

Identity, definitions, proposals and replay without a computer

ACTIVITY	CORE CHALLENGE
IDENTITY DETECTIVE	Give two similar objects different ID cards and histories. Move them to the same place. Ask which facts still distinguish them.
DEFINITION CHAIN	Create three rule cards where C depends on B and B depends on A. Try the wrong order, then topologically sort the cards.
PROPOSAL COUNCIL	Two players propose conflicting changes. Use time, priority or a conflict card. Record accepted and rejected reasons.
REPLAY BEADS	Start from a checkpoint object and replay bead-coded changes. Insert one wrong bead and find the first divergence.

ACCESS ROUTES

Use objects, raised symbols, audio labels, switch scanning, gaze choice or a partner who presents every option faithfully. Offer undo, stop and changed-mind repair.

EVIDENCE NOTE

Success in one activity shows learning in that context. It is not a clinical, cognitive or legal assessment.

FORMAL BRIDGE

Builder contracts for identity and authority

RECORD	MINIMUM MEANING
DefinitionSpec	id, kind, domain, codomain, parameters, dependencies, phase, invariants, provenance, content hash
Instance / Node	stable ID, definition or asset reference, local state, parent, tags and lineage reference
Proposal	event time/type/entity/payload/source/priority/confidence/guard/error/margin/lineage label
Verifier	support + compatibility + guard + confidence/error + policy; emits accept or reason-coded reject
Commit	deterministic order, explicit conflict policy and one atomic state patch
Event record	sequence, pre/post hashes, transition identity, route and external novelty
Checkpoint	canonical snapshot and digest at a declared sequence
Replay	load initial/checkpoint state, apply accepted events in order, stop at first mismatch
Observation	source/capture/calibration/support/value/uncertainty/provenance; proposal-only until verified
KSEED	versioned seed-plus-evidence stream with framing, integrity and reconstruction boundary

PURE VERSUS EFFECTFUL

Pure evaluation may be parallel or cached under a declared order. Effectful state mutation must return to proposal, verification, deterministic commit and lineage.

LATEST SHELF CONNECTIONS

How Volume 3 reaches the newly audited releases

RELEASE / BRANCH	VOLUME 3 CONNECTION
3.9.3 KC ATLAS	Observation, stable map identity, route/change queries and replay.
4.0 SPATIAL LEDGER	General capture/uncertainty/verifier/ledger contract; phone work deferred.
4.1 / 4.1.1	Seed-plus-evidence storage, native source handoff and source/phone evidence separation.
4.2 GENERAL	Operator meaning and all result-affecting orders become explicit data.
4.2 GO SOLVER	Exact certificates only after complete search; heuristic larger-board work stays labeled false/exact.
DB-2.0	Child agency and accessible control remain part of how the library is used.

BRANCH RULE

These releases share the authority discipline but do not form one replacement ladder. The shelf update records dates and parent relationships separately.

SOURCE REGISTER AND EVIDENCE BOUNDARY

Primary technical basis for Volume 3

SOURCE	USED FOR
KC Two Hands 3.0	stable scene/node identity, asset/node split, proposals, conflict policy, checkpoints and replay
UGTS-KC 3.6	literal definitions, content hashes, dependency closure and operation order
SCLP 3.6.2	finite keys, winding, lineage and compression/reconstruction boundaries
SARA 3.6.3	public/private partition, authorization and zero secret egress
KC Elizabeth 3.9	canonical project/snapshot/hash/save and fixed-step event phases
KC Atlas 3.9.3	spatial observations, stable map identity, routes, change and publication
UGTS-KC 4.0	spatial evidence ledger, interval uncertainty and Android deferral
UGTS-KC 4.1 / 4.1.1	KSEED and native source handoff, host/device promotion gates
UGTS-KC 4.2	general OperatorSpec and fifteen-phase order calculus
UGTS-KC 4.2 Go Solver	exact finite certificate versus heuristic search boundary

NO LEGAL OR SAFETY PROMOTION

Hashes, requester labels, technical reports and internal tests do not establish ownership, authorship priority, chain of custody, safety certification or physical-device accuracy.

VOLUME 3 COMPLETE

One consecutive library volume, not the end of the project

What this volume establishes

- Position, asset, node, object, route and lineage are separated.
- Definitions become typed, content-addressed records with explicit dependencies and phase order.
- Observation and proposal are kept separate from deterministic authoritative commit.
- Conflicts, checkpoints, replay and divergence are taught as inspectable processes.
- The spatial seed/evidence boundary and the latest 4.2 general order layer are integrated without overclaiming.

Next volume in the sequence

VOLUME 4 - MAKING PLAYABLE WORLDS

The next book will move from concepts into applications: scenes and assets, two-hand interaction, vector art, named input, animation, collision, procedural sound, saves and self-contained browser delivery.

EDITION CONTROL

UGTS Library Edition, Volume 3, edition 0.1. This educational layer explains source releases and does not modify their formal version numbers.